

$$\Rightarrow M_B = 2M$$

12 D

Consider the forces on the satellite:

$$\frac{GMm}{r^2} = mr\omega^2 \quad \Rightarrow \quad \omega^2 \propto \frac{1}{r^3}$$

So when r decreases, both gravitational force on the satellite and its angular velocity increase. So A and B are wrong statements.

Consider the various energies associated with an orbiting satellite:

$$KE = +\frac{GMm}{2r}, \quad GPE = -\frac{GMm}{r}$$

So when radius r decreases, KE increases while GPE decrease. So only D is correct.

13 B

$m = 40$